

ANSHIKA BAJPAI

anshikabajpai23@gmail.com | github.com/anshikabajpai23 | [linkedin.com/in/anshikabajpai23/](https://www.linkedin.com/in/anshikabajpai23/) | bajpaianshika.netlify.app/ | +19302043030

SUMMARY

Senior Software Engineer and AI Developer with 4+ years of experience building production backend services, distributed systems, data pipelines, and LLM-powered applications. Strong foundation in object-oriented design, design patterns, CI/CD, testing, code reviews, and scalable system development. Experienced in Python, C++, SQL, FastAPI, Docker, Kubernetes, cloud platforms, RAG, and agentic AI workflows.

EDUCATION

Indiana University Bloomington, Master of Science in Data Science, USA (GPA: 3.94)

May 2026

B. Tech in Computer Science and Engineering, Jaypee Institute of Information Technology, India

May 2021

WORK EXPERIENCE

Machine Learning Research Assistant, Indiana University (AIMed Lab)

Jan 2026 – May 2026

- Engineered **DESS→PD medical image translation pipeline** using **RegGAN** on **NVIDIA A100 GPU**, improving distribution alignment by **37% (FID 260→164)** with **0.055px deformation** and **0% topology violations**, **outperforming CycleGAN baseline with 40× lower deformation**
- Developed end-to-end **computer vision preprocessing pipeline** for cross-modality MRI alignment across 11,000+ image slices on **HPC cluster**; evaluated using **FID, SSIM, and Jacobian determinant metrics**; extending to **diffusion models (DDPM)** to generate synthetic healthy knee MRI images, enabling before-and-after injury comparison for personalized treatment planning.

Machine Learning Engineer Intern, Palo Alto Networks, California

May 2025 - Aug 2025

- Built production RAG and backend service layer on GCP using **Vertex AI Embeddings, Enterprise Search, Redis caching**, and API endpoints for document Q&A serving 10,000+ employees; reduced **latency 75%** and **redundant API calls by 41%**.
- Architected **async document ingestion pipeline** via **GCP Pub/Sub** for real-time PDF/DOCX/HTML processing; applied **BigQuery** content hashing and metadata deduplication to cut duplicate ingestion by **~35%** across **500+ internal documentation sources**
- Engineered **multi-turn conversational agent** with **context compression** and **sliding-window memory**, enabling **20+ turn** coherent sessions; improved answer relevance by **31%** via **chain-of-thought** and **few-shot prompt templates**
- Designed and ran **A/B retrieval evals** across chunking strategies, **semantic chunking** boosted **Precision@3 by 18%** on cybersecurity queries, quantifying the impact of prompt and retrieval changes

Senior Software Engineer, Optum (UnitedHealth Group)

Jun 2021 – Jun 2024

- Refactored legacy backend system using **object-oriented design and modular service patterns**, reducing processing time from **500+ minutes to under 3 minutes** and improving reliability for production workflows.
- Designed and deployed **distributed microservices** for compliance notifications, processing millions of notices and saving **\$500K annually** through a company-wide Paperless Initiative.
- Owned **full SDLC** across design, development, testing, code reviews, CI/CD, production support, and documentation; improved error reporting from **25,000 errors to 20** and maintained 90%+ unit test coverage.

Machine Learning Scientist Intern, Taiyo LLC

Dec 2019 – Apr 2020

- Implemented an end-to-end **ML pipeline for stock market prediction**, including **time series forecasting, hyperparameter tuning across 10+ models**, and improving accuracy by **15%** on 50,000+ records.

SKILLS

Programming Languages: Python, Java, C++, SQL, TypeScript, React, R, Linux/Unix

Software Engineering: Object-Oriented Design, Design Patterns, Distributed Systems, Microservices, REST APIs, Unit Testing, Code Reviews, CI/CD, Source Control, Production Support, AWS, GCP, Azure, Docker, Kubernetes, Airflow, Jenkins, Git, MLflow, Grafana

Backend & Data: FastAPI, PostgreSQL, BigQuery, ETL, Redis, PySpark, Snowflake, Streamlit

AI/ML: RAG, LangChain, LangGraph, CrewAI, MCP, OpenAI API, Claude API, Vertex AI, Prompt Engineering, PyTorch, TensorFlow, scikit-learn, XGBoost, AI, Agentic AI, MCP, GenerativeAI, TensorRT-LLM, n8n, vLLM

PUBLICATIONS

- Bajpai A., D. Cavar, et al**, Improving LLM Reasoning Through Ontology-driven Knowledge Graph and FAISS s: A Comparative Study of Generating Ontologies for Medical RAGs, Midwest Speech and Language Days 2025, Indiana University Bloomington, April 2025.
- Bajpai A., Garg M.**, Hindi Sentiment Analysis on Tweets, 2nd International Conference on Artificial Intelligence: Theory & Applications.

PROJECTS

Incident Severity Auto-Triage with RLHF | LLaMA 3.2 1B, QLoRA, GRPO, TRL, PEFT, Hugging Face Transformers, Streamlit, Python, GPU A100

- Fine-tuned LLaMA 3.2 1B with QLoRA on 124K bug reports; diagnosed and fixed token ID mapping bug in weighted cross-entropy loss and gradient dilution bug, lifting **P0 F1 from 0.05 to 0.41** and **team routing accuracy from 0.51 → 0.82**
- Improved **severity macro-F1 from 0.41 to 0.57** across 5 imbalanced classes via mild oversampling and per-class token weighting; reduced **parse failure rate 17.8% to 12%** by fixing train/inference prompt mismatch
- Implementing **RLHF pipeline with FSDP: SFT → Reward Model → GRPO** for incident severity auto-triage on A100 GPUs

FAQ Agent Evaluation Pipeline | Agentic AI, LangGraph, LangChain, LangSmith, OpenEvals, Qwen, Pytest, Python, Claude API, FAISS

- Developed **agentic RAG system** for insurance claims Q&A with **LangGraph, ReAct agents, multi-agent orchestration**, and **hybrid BM25 + vector retrieval**, routing queries across auto, home, and billing agents; implemented **MCP server** to expose retrieval tools as standardized agent interfaces, achieving 83% correctness / 84% faithfulness on LangSmith-evaluated Q&A pairs.
- Built **LLM evaluation framework** with **LLM-as-judge, RAGAS-style metrics, and LangSmith** tracing to evaluate correctness, faithfulness, and retrieval relevance across 60 **adversarial and multi-hop test cases**; surfaced judge inconsistency and score inflation in 7B models, informing retrieval and prompt iteration cycles.